

# 1 HI-Jass User Manual

## 1.1 Running the application

### 1.1.1 Linux

`./run_hi_jass.sh`

### 1.1.2 Windows

Double-click `run_hi_jass.bat`, or run:

`.\run_hi_jass.bat`

The launcher creates the virtual environment, installs the dependencies, and starts the GUI.

## 1.2 Default NBI configuration

The default configuration contains two neutral beams:

| Beam  | Species | Injection energy | Injected power |
|-------|---------|------------------|----------------|
| NBI-1 | D       | 120 keV          | 5 MW           |
| NBI-2 | T       | 180 keV          | 5 MW           |

## 1.3 Two-beam slowing-down model

The slowing-down quantities are calculated independently for each beam. For beam  $j$ :

$$E_{c,j} = \left( \frac{m_j}{m_e} \right)^{1/3} T_e$$

$$\tau_{se,j} = 6.27 \times 10^8 \frac{A_j T_e [\text{eV}]^{3/2}}{n_e [\text{cm}^{-3}] \ln \Lambda}$$

$$\tau_{S,j} = \frac{\tau_{se,j}}{3} \ln \left[ 1 + \left( \frac{E_{b,j}}{E_{c,j}} \right)^{3/2} \right]$$

The useful beam power includes the HotJass capture and loss cascade. The fast-ion density from each beam is:

$$n_{b0,j} = \frac{P_{\text{useful},j} \tau_{S,j}}{E_{b,j} (10^3 e) V}$$

The total fast-ion density is:

$$n_{b0} = \sum_j n_{b0,j}$$

## 1.4 Density peaking and shine-through

The density peaking input defines the analytic electron-density profile:

$$n_e(\rho) = n_{e0}(1 - \rho^2)^{2p_n}, \quad 0 \leq \rho \leq 1$$

Shine-through uses the density along the beam chord rather than assuming that the central density applies everywhere. Each NBI supplies a tangent point  $(R_t, Z_t)$  in the NBI tab. The straight tangential chord at that point is intersected with the shaped plasma cross-section, and the optical depth is evaluated as:

$$\tau_b = \sigma_{\text{stop}}(E_b/A_b) \int_{\text{chord}} n_e(\rho(s)) ds$$

and the captured and shine-through fractions are:

$$f_{\text{capt},b} = 1 - e^{-\tau_b}, \quad f_{\text{shine},b} = e^{-\tau_b}.$$

The chord is evaluated numerically using the local cylindrical radius along the tangent path and the plasma elongation. With  $p_n = 0$ , the density is uniform along the computed chord; with  $p_n > 0$ , the profile is integrated point by point. A tangent point outside the plasma produces zero captured power.

### 1.4.1 Shine-through model selection

Each NBI has a three-way shine-through selector:

- **Riviere:** the existing Riviere-type analytic stopping cross-section fit.
- **Janev/Suzuki:** the Janev-Boley-Post analytic fit from the supplied paper, Nuclear Fusion 29 (1989) 2125, Eqs. (23)-(26), using the published Table III coefficients and the carbon-impurity coefficients from Table IV. It models total effective stopping, including the paper's density, electron-temperature and  $Z_{\text{eff}}$  dependence. The fit is published for  $100 \leq E \leq 10^4$  keV/u,  $10^{12} \leq n_e \leq 10^{15}$  cm<sup>-3</sup> and  $1 \leq T_e \leq 50$  keV; inputs outside these ranges are clipped to the nearest validity boundary.
- **Manual:** the entered manual shine-through fraction is used directly; captured power is  $(1 - f_{\text{shine}})P_{NB}$ .

The selected calculated stopping model is also used by the optional first-orbit-loss attenuation weighting. The Manual mode bypasses the cross-section calculation for shine-through.

## 1.5 First-orbit loss (reduced model)

First-orbit loss is an **opt-in** channel (the *First-orbit loss* checkbox in the Models section). When enabled, it estimates the fraction of *captured* fast ions that leave the plasma promptly on their first drift orbit, before any slowing-down or charge-exchange because they are born close enough to the last closed flux surface (LCFS) that their guiding-centre orbit, or their gyro-orbit, crosses it.

The **Orbit model** selector (Models section) chooses how the orbit widths and the loss criterion are built:

- **Large-aspect** ( $q_*\rho_{Li}$ ) the original reduced model described in *Passing-orbit width* and *Loss criterion* below. Direction-only mean shift; exactly zero co-current loss.
- **ST orbits mean-shift (arbitrary A)** and **ST orbits pitch-resolved** two spherical-tokamak variants described in *ST orbit models* at the end of this section. Both give non-zero co-current loss.

### 1.5.1 Birth profile along the chord

The calculation reuses the shine-through deposition. Along the straight tangential chord (assumed  $R_t = R_0$ , length  $2a$ ) the local ionisation rate density is

$$\dot{n}(x) \propto n_{e0} \sigma_{\text{stop}}(E_{b,j}/A_j) e^{-n_{e0}\sigma_{\text{stop}}x}, \quad x \in [0, 2a],$$

and the path coordinate is mapped to a normalised flux label by

$$\rho(x) = \frac{|x - a|}{a},$$

so  $\rho = 0$  at the chord mid-point and  $\rho = 1$  at both ends. The stopping cross-section is the beam's selected shine-through model; **Manual** mode has no deposition shape, so the Riviere fit is substituted for this weighting only.

### 1.5.2 Passing-orbit width

The radial width of a strongly co-passing drift orbit at the full injection energy is the textbook large-aspect-ratio estimate

$$\Delta r_j = q_* \rho_{Li,j}, \quad q_* = \frac{2\pi a^2 B_t}{\mu_0 R_0 I_p} \frac{1 + \kappa^2}{2}, \quad \rho_{Li,j} = \frac{m_j v_{b,j}}{e B_t}.$$

$q_*$  is the cylindrical, elongation-corrected safety factor and  $\rho_{Li,j}$  the toroidal-field gyroradius at  $E_{b,j}$ .

### 1.5.3 Loss criterion

The orbit is modelled as a **rigid mean shift** of the birth radius: inward for co-current injection (favourable), outward for counter-current:

$$\rho_{\text{eff}}(x) = \rho(x) \mp \frac{\Delta r_j}{a} \quad (\text{co / counter}),$$

and an ion born at  $x$  is counted as lost if  $\rho_{\text{eff}}(x) > 1$ . Because  $\rho(x) \leq 1$ , the co-current shift can never trigger this test, so the mean-shift model gives **exactly zero first-orbit loss for co-current injection**; for counter-current it loses every ion born with  $\rho(x) > 1 - \Delta r_j/a$ .

An optional, direction-independent **Larmor (gyro-orbit) prompt-loss** channel can be OR-ed in: an ion whose guiding centre is born within one gyro-diameter of the LCFS,

$$\rho(x) > 1 - \frac{2\rho_{Li,j}}{a},$$

is lost regardless of the drift direction (gyration is much faster than the guiding-centre drift). This channel is disabled by default.

#### 1.5.4 Loss fraction and power

$$f_{\text{orbit},j} = \frac{\int_{\text{lost}}^x \dot{n}(x) dx}{\int_0^{2a} \dot{n}(x) dx}, \quad P_{\text{orbit},j} = f_{\text{orbit},j} P_{\text{capt},j}.$$

First-orbit loss is applied to the captured power *before* the charge-exchange loss and before the useful power that drives  $T_e$ ,  $T_i$  and  $n_{b0,j}$ . Injection direction (co-/counter-current) is set per NBI in its input section.

#### 1.5.5 Reported quantities

The Dashboard and the Assumptions tab report, per beam,  $\rho_{Li,j}$  and  $\Delta r_j$  (in cm and as fractions of  $a$ ) and  $f_{\text{orbit},j}$ .

#### 1.5.6 Limitations

- **The co-current zero is a model artefact, not physics.** The rigid mean-shift ignores the finite radial width of the orbit and the outboard excursion that a favourable inward mean shift does not cancel; published orbit-following comparisons give of order a few percent co-current first-orbit loss, versus tens of percent counter-current.
- The chord label  $\rho(x) = |x - a|/a$  folds the inboard and outboard halves together, so the model cannot distinguish a high-field-side birth from a low-field-side birth. It therefore cannot represent the correct structure co-current losing high-field-side edge births, counter-current losing low-field-side edge births, and trapped-ion banana-tip losses on the low-field side in either direction.
- Circular cross-section, large-aspect-ratio,  $R_t = R_0$  tangential geometry;  $\Delta r_j$  and  $\rho_{Li,j}$  are evaluated at the full injection energy (an upper bound real ions slow down).
- The birth-profile integral in this channel uses the flat central density  $n_{e0}$ , not the peaked profile used elsewhere.
- When  $\rho_{Li}/a$  or  $\Delta r/a \gtrsim 0.3$  orbit width comparable to the machine size, typical of compact low-aspect-ratio devices the whole estimate is order-unity uncertain and really needs a guiding-centre orbit follower. The Assumptions tab flags this case. The ST orbit models below address the first three of these limitations directly.

#### 1.5.7 ST orbit models (arbitrary aspect ratio)

On a spherical tokamak the drift and banana orbit widths are an  $O(a)$  fraction of the minor radius, the trapped fraction is large, and outboard-born trapped ions are lost **regardless of injection direction** so the large-aspect model's exact co-current zero is qualitatively wrong. The two ST variants build their widths from the **poloidal gyroradius**

$$\rho_{\theta,j} = \frac{B_t}{\bar{B}_p} \rho_{Li,j}, \quad \bar{B}_p = \frac{\mu_0 I_p}{2\pi a \sqrt{(1 + \kappa^2)/2}},$$

(not  $q_* \rho_{Li}$ , whose  $q_*$  runs away as  $\varepsilon \rightarrow 1$ ), together with the circulating and banana widths and the arbitrary- $A$  trapped fraction

$$w_{\text{pass},j} = \varepsilon \rho_{\theta,j}, \quad w_{\text{ban},j} = 2\sqrt{\varepsilon} \rho_{\theta,j}, \quad f_t = 1 - \frac{(1 - \varepsilon)^2}{(1 + 1.46\sqrt{\varepsilon})\sqrt{1 - \varepsilon^2}},$$

with  $\varepsilon = a/R_0$  (clamped to  $[10^{-3}, 0.95]$ ) and  $f_c = 1 - f_t$  (Lin-Liu & Miller, *Phys. Plasmas* **2** (1995) 1666). An arbitrary- $A$  edge safety factor  $q_a$  (Uckan / ITER Physics Basis form, with the  $(1.17 - 0.65\varepsilon)/(1 - \varepsilon^2)^2$  shaping factor) is reported for reference but is **not** used in the widths.

The birth profile in this channel is rebuilt from a physically calibrated mean free path  $\lambda_{\text{mfp}} \approx 5.5 \times 10^{19} (E_b/A)/n_{e0}$  m, clamped to  $[0.2a, 8a]$ , instead of the (with *Manual* shine-through, pathologically edge-peaked) Riviere weighting.

The **drift-orbit shift is direction-dependent** for both populations inward for co-current ( $v_{\parallel}$  along  $I_p$ ), outward for counter with a weaker shift  $c_t w_{\text{pass}}$  ( $c_t = 0.25$ ) for the trapped-ion banana centre than for a passing ion. The **gyro channel** ( $\rho > 1 - \rho_{Li}/a$ , born within one gyroradius of the LCFS) is direction-independent. Co-current loss is therefore non-zero but smaller than counter-current the co/counter asymmetry survives, unlike the large-aspect model's all-or-nothing 0 vs 1.

**Mean-shift variant.** Deposition-weighted loss integrals over  $\rho(x)$ , blended by  $f_c, f_t$  (sign convention:  $-$  co,  $+$  counter):

$$f_{\text{orbit},j} = f_c L_{\text{pass}} + f_t L_{\text{trap}},$$

$$L_{\text{pass}} : \rho \mp \frac{w_{\text{pass},j}}{a} > 1 \quad \text{or} \quad \rho > 1 - \frac{\rho_{Li,j}}{a},$$

$$L_{\text{trap}} : \rho + \frac{w_{\text{ban},j}}{2a} \mp \frac{0.25 w_{\text{pass},j}}{a} > 1 \quad \text{or} \quad \rho > 1 - \frac{\rho_{Li,j}}{a}.$$

**Pitch-resolved variant.** A 2-D integral over birth radius  $x$  and the pitch **magnitude**  $|\lambda| = |v_{\parallel}/v|$ . The birth value  $|\lambda_0(x)| = R_t/\sqrt{R_t^2 + (x-a)^2}$  is fixed by the injection *geometry* and does **not** depend on the poloidal-field / current direction only the sign of  $v_{\parallel}$  relative to  $I_p$  flips (co  $\leftrightarrow$  counter), carried by  $s_{\text{dir}} = \mp 1$ .  $|\lambda_0|$  is smeared by a  $\sigma = 0.10$  Gaussian. An ion is trapped if  $|\lambda| < \sqrt{2\varepsilon_{\text{loc}}/(1 + \varepsilon_{\text{loc}})}$  ( $\varepsilon_{\text{loc}} = \rho a/R_0$ ); its outboard radial excursion is

$$\text{passing} : s_{\text{dir}} \frac{w_{\text{pass},j}}{a} |\lambda|, \quad \text{trapped} : \frac{w_{\text{ban},j}}{2a} \frac{|\lambda|}{\lambda_{\text{tr}}} + s_{\text{dir}} \frac{0.25 w_{\text{pass},j}}{a},$$

(the banana half-width  $\rightarrow 0$  for deeply trapped ions, largest near the trapped/passing boundary), and it is lost if  $\rho + (\text{excursion}) > 1$  or  $\rho > 1 - \rho_{Li,j}/a$ .

Both variants collapse toward the large-aspect co-current zero as  $\varepsilon \rightarrow 0$ . They remain 0-D order-of-magnitude estimates absolute magnitudes are upper-bound-like and, because the direction-independent trapped banana-tip and gyro channels stay significant, the co/counter split is narrower than the large-aspect model's (typically a factor  $\sim 1.5$ , not 0 vs 1). References: Akers *et al.*, *Nucl. Fusion* (START NBI,  $A \sim 1.4$ ); Goldston, White & Boozer, *Phys. Rev. Lett.* **47** (1981) 1004; Goldston & Rutherford, *Introduction to Plasma Physics* (1995), Ch. 12.

## 1.6 Power balance (electron and ion)

At each central density the model solves a 0-D steady-state power balance for  $T_e$  and  $T_i$ . The heating that drives each temperature is the sum of the neutral-beam split, prescribed auxiliary heating (ECRH and ICRH), and when enabled fusion alpha self-heating.

### 1.6.1 Beam heating split

Each useful beam  $j$  delivers a fraction  $L_e(x_j)$  of its power to electrons and  $L_i(x_j) = 1 - L_e(x_j)$  to ions, with  $x_j = E_{b,j}/E_{c,j}(T_e)$  and

$$L_i(x) = \frac{1}{x} \int_0^x \frac{du}{1+u^{3/2}}, \quad L_e(x) = 1 - L_i(x).$$

$L_i \rightarrow 1$  as  $x \rightarrow 0$  (cold, ion-dominated) and  $L_i \rightarrow 0$  as  $x \rightarrow \infty$  (fast beam, electron-dominated).  $P_{\text{useful},j}$  is the injected power after shine-through, first-orbit loss and charge-exchange loss (preceding sections).

### 1.6.2 Auxiliary heating (ECRH, ICRH)

ECRH and ICRH are entered as powers with electron/ion fractions. Their contribution to each channel is

$$P_{\text{aux},e} = f_e^{\text{EC}} P_{\text{ECRH}} + f_e^{\text{IC}} P_{\text{ICRH}}, \quad P_{\text{aux},i} = (1 - f_e^{\text{EC}}) P_{\text{ECRH}} + f_i^{\text{IC}} P_{\text{ICRH}}.$$

ECRH is electron-dominant,  $f_e^{\text{EC}}$  close to 1. For ICRH the electron and ion fractions are set independently; if  $f_e^{\text{IC}} + f_i^{\text{IC}} < 1$  the remainder is treated as power that does not thermalise (a fast-ion tail or direct loss). These sources are Te/Ti-independent constants added to the right-hand side of each balance.

### 1.6.3 Alpha self-heating (optional)

When enabled, 3.5 MeV fusion alphas deposit

$$P_\alpha = f_\alpha \frac{E_\alpha}{E_f} P_{f,\text{tot}} = f_\alpha \frac{3.5}{17.6} P_{f,\text{tot}},$$

split between electrons and ions by the alpha slowing-down fraction

$$L_e^\alpha = L_e \left( \frac{E_\alpha}{E_c^\alpha(T_e)} \right), \quad E_c^\alpha = \left( \frac{m_\alpha}{m_e} \right)^{1/3} T_e,$$

so  $P_{\alpha,e} = L_e^\alpha P_\alpha$  and  $P_{\alpha,i} = (1 - L_e^\alpha) P_\alpha$ . Because  $P_\alpha$  depends on  $P_{f,\text{tot}}$ , which depends on  $T_i$ , which depends on  $P_{\alpha,i}$ , the alpha term is closed by an under-relaxed fixed-point iteration; with alpha heating off the solve is a single pass.

### 1.6.4 Decoupled balance (default)

With electron-ion equipartition off, the two balances are solved in sequence. The electron balance is a 1-D root-find for  $T_e$ :

$$\frac{3}{2} n_{e0} T_e (10^3 e) \frac{V}{\tau_{E,e}} = \sum_j L_e(x_j) P_{\text{useful},j} + P_{\text{aux},e} + P_{\alpha,e}.$$

The ion temperature then follows in closed form (or, for the neoclassical-ion option, from a second 1-D root-find see *Confinement time*):

$$\frac{3}{2} n_{\text{thermal}} T_i (10^3 e) \frac{V}{\tau_{E,i}} = \sum_j L_i(x_j) P_{\text{useful},j} + P_{\text{aux},i} + P_{\alpha,i},$$

where  $n_{\text{thermal}} = n_{\text{sum}}(n_{e0}, Z_{\text{eff}}) - n_{b0}$  is the thermal-ion density left after the fast ions are subtracted. A non-positive  $n_{\text{thermal}}$  marks the operating point infeasible.

### 1.6.5 Coupled balance (equipartition on)

When electron-ion equipartition is enabled, the exchange power  $P_{ei}(T_e, T_i)$  (defined in *Electron-ion exchange time*) is subtracted from the electron balance and added to the ion balance, and the pair is solved together:

$$\begin{aligned} \frac{3}{2} n_{e0} T_e (10^3 e) \frac{V}{\tau_{E,e}} &= \sum_j L_e(x_j) P_{\text{useful},j} + P_{\text{aux},e} + P_{\alpha,e} - P_{ei}(T_e, T_i), \\ \frac{3}{2} n_{\text{thermal}} T_i (10^3 e) \frac{V}{\tau_{E,i}} &= \sum_j L_i(x_j) P_{\text{useful},j} + P_{\text{aux},i} + P_{\alpha,i} + P_{ei}(T_e, T_i). \end{aligned}$$

The reported  $P_e$  and  $P_i$  remain the pure beam split;  $P_{ei}$ ,  $P_{\alpha}$  and  $P_{\text{aux}}$  are reported separately.

## 1.7 Confinement time

$\tau_{E,e}$  and  $\tau_{E,i}$  in the balances above are set per channel by the *Confinement* selector in the Models section; the electron and ion channels are chosen independently. When a physics-based scaling is selected the manual `tauE,e` / `tauE,i` inputs are ignored, and the Summary sheet reports the computed value (or its range across a scan) instead.

The physics-based scalings below all use the same loss power

$$P_{\text{loss}} = \sum_j P_{\text{useful},j} + P_{\text{aux},e} + P_{\text{aux},i},$$

i.e. the captured, post-orbit, post-charge-exchange beam power plus the prescribed auxiliary (ECRH, ICRH) heating. Fusion alpha heating is not included in  $P_{\text{loss}}$  (it is closed by an outer fixed-point and is not known when  $\tau_E$  is evaluated). None of these depend on  $T_e$  or  $T_i$ , so each is evaluated once.

### 1.7.1 Fixed (input)

$\tau_{E,e}$  and  $\tau_{E,i}$  are the values entered on the Plasma tab, used unchanged.

### 1.7.2 IPB98(y,2) ELMy H-mode

The ITER Physics Basis ELMy H-mode scaling (ITER Physics Basis, *Nucl. Fusion* **39** (1999) 2175; standard reference form):

$$\tau_E = 0.0562 I_p^{0.93} B_t^{0.15} n_{19}^{0.41} P_{\text{loss}}^{-0.69} R_0^{1.97} \kappa_a^{0.78} \varepsilon^{0.58} M_{\text{eff}}^{0.19}$$

with  $I_p$  in MA,  $B_t$  in T,  $P_{\text{loss}}$  in MW,  $R_0$  and  $a$  in m,  $n_{19} = n_{e0}/10^{19} \text{ m}^{-3}$ ,  $\varepsilon = a/R_0$ ,  $\kappa_a = \kappa$ , and  $M_{\text{eff}} = 2x_D + 3x_T$  the mass-weighted ion mass number. Selecting *IPB98(y,2) ELMy H-mode* uses

it for both channels ( $\tau_{E,e} = \tau_{E,i}$ ); *IPB98(y,2) e / neoclassical i* uses it for the electron channel only, with the ion channel from the neoclassical estimate below.

The fit was obtained from conventional-aspect-ratio devices ( $A \sim 2.5$ -4) and is documented in the spherical-tokamak literature to mis-predict at low aspect ratio. When  $A = R_0/a < 2$  the Assumptions tab flags that IPB98(y,2) is being extrapolated outside its dataset.

### 1.7.3 Kaye NSTX L-mode

An NSTX low-aspect-ratio (spherical tokamak) L-mode global fit (Kaye, *Nucl. Fusion* **46** (2006) 848, Eq. 5 / Fig. 9):

$$\tau_E = 4.73 \times 10^{-4} I_p^{1.01} B_t^{0.70} n_e^{-0.07} P_{\text{loss}}^{-0.37}$$

with  $I_p$  in A,  $B_t$  in T,  $n_e$  in  $\text{m}^{-3}$ ,  $P_{\text{loss}}$  in W and  $\tau_E$  in s. Selecting *Kaye NSTX L-mode* uses it for both channels ( $\tau_{E,e} = \tau_{E,i}$ ); *Kaye L e / neoclassical i* uses it for the electron channel only.

### 1.7.4 Kaye NSTX H-mode

The H-mode counterpart of the previous fit: an NSTX low-aspect-ratio (spherical tokamak) H-mode thermal-energy scaling (Kaye, *Nucl. Fusion* **46** (2006) 848, Table 1, ordinary-least-squares "Case 1" fit to all 85 H-mode points, RMSE = 0.145):

$$\tau_E = 4.69 \times 10^{-9} I_p^{0.57} B_t^{1.08} n_e^{0.44} P_{\text{loss}}^{-0.73}$$

with  $I_p$  in A,  $B_t$  in T,  $n_e$  in  $\text{m}^{-3}$ ,  $P_{\text{loss}}$  in W and  $\tau_E$  in s. Relative to IPB98(y,2) the current dependence is much weaker ( $I_p^{0.57}$  vs  $I_p^{0.93}$ ) and the toroidal-field dependence much stronger ( $B_t^{1.08}$  vs  $B_t^{0.15}$ ), the qualitative spherical-tokamak trend; the density and heating-power exponents ( $n_e^{0.44}$ ,  $P_{\text{loss}}^{-0.73}$ ) are of the same order as the conventional-aspect scalings. This is an ST-appropriate fit ( $A \sim 1.31.5$  dataset). Selecting *Kaye NSTX H-mode* uses it for both channels ( $\tau_{E,e} = \tau_{E,i}$ ); *Kaye H e / neoclassical i* uses it for the electron channel only.

### 1.7.5 Neoclassical ion

A banana-regime order-of-magnitude estimate (Wesson, *Tokamaks*; Chang-Hinton low-collisionality limit) for the ion channel only:

$$\chi_{i,\text{neo}} \simeq \frac{q^2 \rho_i^2 \nu_{ii}}{\varepsilon^{3/2}}, \quad \tau_{E,i} = \frac{a^2}{\chi_{i,\text{neo}}} = \frac{a^2 \varepsilon^{3/2}}{q^2 \rho_i^2 \nu_{ii}},$$

with  $\varepsilon = a/R_0$ , the elongation-corrected cylindrical safety factor

$$q = \frac{5 a^2 B_t}{R_0 I_p [\text{MA}]} \frac{1 + \kappa^2}{2},$$

the thermal-ion Larmor radius

$$\rho_i = \frac{m_{\text{eff}} v_{th,i}}{e B_t}, \quad v_{th,i} = \sqrt{\frac{2 T_i (10^3 e)}{m_{\text{eff}}}}, \quad m_{\text{eff}} = x_D m_D + x_T m_T,$$

and the ion-ion collision frequency summed over D and T,

$$\nu_{ii} = \sum_{s \in \{D, T\}} 4.80 \times 10^{-8} \frac{n_s [\text{cm}^{-3}] \ln \Lambda}{\sqrt{\mu_s} T_i [\text{eV}]^{3/2}}, \quad \mu_s = m_s / m_p.$$

Since  $\tau_{E,i} \propto T_i^{1/2}$ , the ion balance becomes a 1-D fixed-point root-find for  $T_i$  rather than a closed-form substitution. Pure neoclassical transport has no anomalous channel, so it under-estimates real ion transport and can predict very hot, sometimes implausible  $T_i$ ; treat it as a lower bound. It cannot be combined with electron-ion equipartition.

### 1.7.6 Neoclassical ion (arbitrary aspect ratio)

The bare  $\varepsilon^{-3/2}$  geometric factor above both diverges as  $\varepsilon \rightarrow 0$  and stays bounded as  $\varepsilon \rightarrow 1$ , neither of which is right on a spherical tokamak. The *arbitrary A* variant (selector entries ending "neoclassical i (arbitrary A)") replaces the trapped-fraction part with the Lin-Liu & Miller  $f_t/f_c$  the banana-regime structure of Helander & Sigmar's *Collisional Transport in Magnetized Plasmas*, Ch. 11 and uses the arbitrary- $A$  edge safety factor  $q_a$ :

$$\chi_{i,\text{neo}} \simeq q_a^2 \rho_i^2 \nu_{ii} \frac{f_t}{f_c}, \quad f_t = 1 - \frac{(1 - \varepsilon)^2}{(1 + 1.46\sqrt{\varepsilon})\sqrt{1 - \varepsilon^2}}, \quad f_c = 1 - f_t,$$

$$q_a = \frac{5 a^2 B_t}{R_0 I_p [\text{MA}]} \frac{1 + \kappa^2(1 + 2\delta^2 - 1.2\delta^3)}{2} \frac{1.17 - 0.65 \varepsilon}{(1 - \varepsilon^2)^2}.$$

$f_t/f_c$  stays finite for every  $\varepsilon < 1$  and rises steeply only as  $\varepsilon \rightarrow 1$  (all ions trapped) the physically correct low-aspect trend. This is **not** a smooth reduction of the  $\varepsilon^{-3/2}$  form: the two order-of-magnitude estimates can differ by up to about one order of magnitude in either direction (more once the  $\tau_{E,i} \propto T_i^{1/2}$  fixed point amplifies it), which is exactly why both are offered run them side by side. References: Helander, *Phys. Plasmas* **7** (2000) 3999; Hinton, Wiley *et al.*, *Phys. Rev. Lett.* **29** (1972) 698; Satake *et al.*, *Phys. Plasmas* **9** (2002); Goldston & Rutherford (1995).

## 1.8 Mean fast-ion energy

Each beam has its own steady-state slowing-down distribution:

$$f_j(E) \propto \frac{\sqrt{E}}{E^{3/2} + E_{c,j}^{3/2}}, \quad 0 < E \leq E_{b,j}$$

Its mean fast-ion energy is:

$$\langle E_{fast} \rangle_j = \frac{\int_0^{E_{b,j}} E f_j(E) dE}{\int_0^{E_{b,j}} f_j(E) dE}$$

## 1.9 Combined values shown in Results

The GUI reports power-weighted effective values for the two beams.

Effective slowing-down time:

$$\tau_S^{\text{eff}} = \frac{\sum_j P_{\text{useful},j} \tau_{S,j}}{\sum_j P_{\text{useful},j}}$$

Effective mean fast-ion energy:

$$\langle E_{\text{fast}} \rangle = \frac{\sum_j P_{\text{useful},j} \langle E_{\text{fast}} \rangle_j}{\sum_j P_{\text{useful},j}}$$

These effective values are used for the combined fast-ion diagnostics, including fast-ion energy density, pressure, beta, and:

$$R = \frac{u_{\text{fast}}}{U_t}$$

The underlying HotJass calculation still evaluates each beam separately before forming these combined diagnostics.

### 1.10 Results tabs (Operating point)

- **Dashboard** scalar table of the operating point (temperatures, densities, power balance terms, orbit widths, Greenwald ratio).
- **Deposition** (1) beam targeting geometry in the torus top view, tangent to each beam's  $R_t$  with its co-/counter-current sense; (2) neutral-beam survival  $I(s)/I_0$  and the fast-ion birth rate  $n_e \sigma I(s)/I_0$  along the beam; (3) the resulting fast-ion birth versus normalised radius  $\rho$ , per beam and total (chord-sampled, midplane circular approximation); (4) the steady-state slowing-down distribution  $f(E)$  with the  $E_b$  edge marked.
- **Power flow** waterfall / Sankey / pie of injected power to its sinks.
- **Profiles**  $n_e(\rho)$ ,  $T_{e,i}(\rho)$  and the plasma shape.
- **Assumptions** the full validity / fit-range read-out and any warnings.
- **References** the literature behind the *currently selected* models (beam stopping, confinement scaling, orbit-loss model) and the active machine geometry, each with a link, plus a feedback button.

### 1.11 Electron-ion exchange time

The electron-ion exchange time is based on the NRL Plasma Formulary thermal-equilibration rate. For each thermal ion species  $s$  (D or T):

$$\bar{\nu}_{ei,s} = 1.8 \times 10^{-19} \frac{\sqrt{m_e m_s} Z_e^2 Z_s^2 n_e \ln \Lambda}{(m_s T_e + m_e T_i)^{3/2}}$$

In this practical-unit expression, masses are in grams, temperatures are in eV, and electron density is in  $\text{cm}^{-3}$ . The model uses  $Z_e = Z_s = 1$  for hydrogenic D and T ions.

For a D-T plasma, the effective exchange time is:

$$\tau_{IE} = \frac{n_D + n_T}{n_D \bar{\nu}_{ei,D} + n_T \bar{\nu}_{ei,T}}$$

For a single thermal ion species, this reduces to:

$$IE = \frac{1}{\bar{\nu}_{ei}}$$

The equivalent energy-over-power form used by the application is:

$$IE = \frac{\frac{3}{2}(n_D + n_T)|T_i - T_e|(10^3 e)V}{|P_{ie}|}$$

where  $V$  is the plasma volume and  $e = 1.602176634 \times 10^{-19}$  J/eV. The exchange power is:

$$P_{ie} = -P_{ei} = -\frac{3}{2}V(T_e - T_i)(n_D\bar{\nu}_{ei,D} + n_T\bar{\nu}_{ei,T})(10^3 e)$$

The absolute values make  $\tau_{IE}$  positive regardless of whether electrons heat ions ( $T_e > T_i$ ) or ions heat electrons ( $T_i > T_e$ ).

## 1.12 Fusion power

The total fusion power is the sum of the thermal D-T and beam-target contributions:

$$P_f = P_{f,\text{thermal}} + P_{f,\text{beam}}$$

### 1.12.1 Thermal D-T fusion

The thermal contribution is calculated from the thermal deuterium and tritium densities and the ion-temperature-dependent Bosch-Hale reactivity:

$$P_{f,\text{thermal}} = n_{D0}n_{T0}\langle\sigma v\rangle_{DT}(T_i)VE_f$$

Here  $n_{D0}$  and  $n_{T0}$  are the thermal species densities,  $V$  is the plasma volume, and:

$$E_f = 17.6 \text{ MeV} = 17.6 \times 10^6 e$$

The Bosch-Hale thermal reactivity is evaluated at  $T_i$ , because the relative velocity is that of the thermal ions, not the electrons.

### 1.12.2 Beam-target fusion

Each useful beam is evaluated separately against the stationary thermal ions of the other D-T species:

$$P_{f,\text{beam}} = \sum_j VE_f \int_0^{E_{b,j}} n_{\text{target},j} f_j(E) [\sigma v]_j(E) dE$$

The beam slowing-down distribution is normalized to the fast-ion density of that beam:

$$f_j(E) \propto \frac{\sqrt{E}}{E^{3/2} + E_{c,j}^{3/2}}, \quad \int_0^{E_{b,j}} f_j(E) dE = n_{b0,j}$$

The beam-target reactivity is:

$$[\sigma v]_j(E) = \sigma_{DT}(E)v_j(E), \quad v_j(E) = \sqrt{\frac{2E}{m_j}}$$

For a D beam,  $n_{\text{target},j} = n_{T0}$ . For a T beam,  $n_{\text{target},j} = n_{D0}$ . The beam fast-ion density is obtained from the useful beam power:

$$n_{b0,j} = \frac{P_{\text{useful},j} \tau_{S,j}}{E_{b,j}(10^3 e)V}$$

The input shine-through fraction is used to determine captured beam power; the plotted  $P_{\text{shine}}$  is then the derived shine-through power from the scan. Charge-exchange and other configured losses similarly reduce the useful power before calculating  $n_{b0,j}$ .

When profile peaking is nonzero, the fusion post-processing uses volume integrals. The thermal contribution becomes:

$$P_{f,\text{thermal}} = V E_f \langle n_D(\rho) n_T(\rho) \langle \sigma v \rangle_{DT} [T_i(\rho)] \rangle_V.$$

The beam-target contribution similarly uses the local target density and the local slowing-down distribution evaluated with  $T_e(\rho)$ . Pressure and thermal-energy diagnostics use the corresponding volume-averaged  $nT$  profiles. The electron and ion power-balance solve remains a 0D central-value solve; peaking changes those temperatures only indirectly through the profile-aware shine-through and captured beam power.

### 1.12.3 Current model limitations

- The beam-target integral uses stationary target ions, so the solved  $T_i$  is not included in that reaction integral.
- D-D beam-target reactions are not included.
- Each included D-T reaction is assigned 17.6 MeV of fusion energy.
- The radial volume weighting uses the model's elliptical flux-surface approximation and does not yet solve a 2D equilibrium.

### 1.13 Density scan

The Plasma tab defines the requested scan range with `n_e_min` and `n_e_max`. The Results tab reports the physically valid interval separately. Points below the HotJass charge-neutrality feasibility limit are not assigned physical output values and are omitted from plots.

### 1.14 Important notation

- $T_e$ : electron temperature
- $T_i$ : thermal-ion temperature
- $P_e$ : beam power deposited into electrons
- $P_i$ : beam power deposited into ions
- $L_e, L_i$ : beam-power electron / ion heating fractions ( $L_e + L_i = 1$ )
- $P_{ie}$ : electron-ion energy exchange power
- $P_{\text{ECRH}}, P_{\text{ICRH}}$ : ECRH and ICRH auxiliary heating power
- $P_{\text{aux},e}, P_{\text{aux},i}$ : auxiliary heating to electrons / ions

- $P_\alpha, P_{\alpha,e}, P_{\alpha,i}$ : fusion alpha self-heating power (total, to electrons, to ions)
- $\tau_{E,e}, \tau_{E,i}$ : electron / ion energy confinement times used in the balance
- $n_{\text{thermal}}$ : thermal-ion density,  $n_{\text{sum}}(n_{e0}, Z_{\text{eff}}) - n_{b0}$
- $n_D, n_T$ : thermal deuterium and tritium densities
- $n_{b0}$ : total fast-ion density
- $\rho_{Li}$ : fast-ion toroidal-field Larmor radius at the injection energy
- $\Delta r$ : passing drift-orbit radial width,  $q_* \rho_{Li}$  (large-aspect model)
- $\rho_\theta$ : fast-ion poloidal gyroradius,  $(B_t/\bar{B}_p) \rho_{Li}$  (ST orbit models)
- $w_{\text{pass}}, w_{\text{ban}}$ : ST circulating / banana orbit widths
- $f_t, f_c$ : trapped / circulating particle fractions (arbitrary  $A$ )
- $q_a$ : arbitrary-aspect-ratio edge safety factor (Uckan form)
- $f_{\text{orbit}}$ : first-orbit loss fraction of captured beam power
- $P_{\text{orbit}}$ : first-orbit loss power
- $P_{f,tot}, P_{f,th}, P_{f,b}$ : total, thermal, and beam-target fusion power
- $\tau_{IE}$ : effective electron-ion exchange time
- $p_{th}, p_{fast}$ : thermal and fast-ion pressure
- $\beta_t$ : toroidal beta